Bridging the Divide: The Effect of Reinforcement Learning from AI Feedback and Supervised Fine-Tuning on the Performance of Smaller Language Models

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Abstract

Reinforcement learning from AI Feedback (RLAIF), introduced by Bai et al. (2022) and developed by Lee et al. (2023), has shown performance in aligning large language models (LLMs) with human preferences that is comparable with reinforcement learning with human feedback (RLHF). Additionally, it does not suffer from the high cost of extracting human preferences which is problematic for RLHF. In this article, we explore the effect of performing RLAIF and supervised fine-tuning (SFT) on small LLMs. We investigate the extent to which this can reduce the performance gap between small LLMs and the largest, state-of-the-art LLMs. To do this, we train small checkpoints of three widely-used LLMs (T5, BART, GPT-2) on a one-sentence summarisation task. Our results indicate that, when measured using ROUGE, smaller models show a significant increase in performance following SFT, but that RLAIF does not result in a further increase. Further, the small models are not able to entirely bridge the gap with the largest state-of-the-art LLMs, but show impressive performance that may encourage organisations with data security concerns to fine-tune models for their specific tasks in-house.

1 Introduction

A range of powerful large language models (LLMs) are now widely available to the public through Application Programming Interfaces. However, these models carry potential privacy and data security concerns and can deter organisations from making wider use of them if they are unable to develop their own. One way around this issue is through the refinement of low-resource model-performance techniques, allowing developers to employ their own models, and removing these concerns. In this project, we explore the effect of tuning small LLMs for text summarisation using supervised fine-tuning (SFT) and reinforcement learning with AI feedback (RLAIF). By using these methods, we aim to investigate to what extent low-resource methods can close the performance gap with larger LLMs and increase public accessibility to model enhancement.

Reinforcement learning (RL) tasks require discovering optimal actions which maximise numerical reward. The primary attributes that distinguish RL problems are their inherent closed-loop nature, absence of explicit instructions for actions, and that decision consequences may unfold over time (Sutton and Barto, 2014). Recently, reinforcement learning with human feedback (RLHF) has been seen as a useful technique for aligning summaries with human preferences. RLHF summaries were in many cases preferred to human-generated summaries, showing the strength of RL for this application (Stiennon et al., 2022). However, RLHF faces issues with scalability due to the high cost of gathering human labels such that this data remains limited.

One solution to this is to replace human feedback with AI feedback. Initially proposed by the team at Anthropic (Bai et al., 2022), this method was designed to allow AI models to improve other AI and develop a harmless AI assistant through self-critique. Following this work, Lee et al. (2023; herein referred to as LEA) developed on RLAIF, producing results comparable to RLHF. Therefore, it is clear that RLAIF boasts impressive utility in enhancing LLM alignment with human preferences. It is upon such RLAIF this project aims to build. However, previous research relies on large-scale language models with billions of parameters. Therefore, it is of interest to investigate smaller-scale models for integrating SFT and RLAIF.

Our results suggest that SFT greatly improves the performance of small models for text summarisation. However, our implementation of RLAIF led to slightly poorer performance than SFT, based on the evaluation metrics used (Section 3.3.4).

The rest of this report proceeds as follows: in Section 2 we review the relevant literature; in Section 3 we describe our methodology; in Sections 4 and 5 we present and discuss our results; and we give final remarks in Section 5.1.

2 Literature Review

Text summarisation is an ever-present task in natural language processing. Traditional extractive techniques, such as PageRank (Brin and Page, 1998), LexRank (Erkan and Radev, 2004), and SumBasic (Nenkova and Vanderwende, 2005), use statistical rules to identify text features and generate summaries from pre-existing tokens, but often overlook linguistic nuances. In recent years, deep neural networks have demonstrated their ability to better incorporate semantic meaning with abstractive summarisation techniques, generating summaries based on key ideas and concepts in the text, which do not necessarily contain pre-existing tokens. For example, several authors have been able to outperform state-of-the-art summarisation systems through abstractive techniques by incorporating LSTM-CNN models (Song et al., 2018) and Convolutional Sequence to Sequence models (Narayan et al., 2018, herein referred to as NCL). Large, off-the-shelf transformer-based LLMs such as PaLM 2 (Dai et al., 2023), ChatGPT, MPT-7b-instruct, and falcon-7b-instruct have also repeatedly been shown to excel at summarisation tasks (Basyal and Sanghvi, 2023).

Summarisation metrics are essential to compare the performance of models such as these. Across summarisation tasks, ROUGE (Recall-Oriented Understudy for Gisting Evaluation) (Lin, 2004) is an intuitive metric based upon N-gram overlap which is the most widely reported. As a result, we use the ROUGE score for comparison with other studies.

Despite boasting impressive general language understanding, LLMs often require further training to excel at specific tasks. This process, known as SFT, involves adjusting the LLM's internal parameters using task-specific data. A large body of research has shown that this technique can produce impressive domain-specific results, such as in financial news summarisation (Baldeon Suarez et al., 2020; Passali et al., 2021) and clinical information extraction (Mulyar and McInnes, 2020).

Many problems that are suitable for SFT can also be addressed using RL. In the context of summarisation, however, several authors have noted the difficulty of aligning a numerical reward model with the human concept of summary quality. To address this, they incorporated human feedback via RLHF and demonstrated a significant improvement in summary quality (Stiennon et al., 2022; Ouyang et al., 2022). However, as mentioned, scaling this methodology is difficult due to the time-consuming and costly nature of human labelling. Recently, studies such as LEA and Bai et al. (2022) have addressed this by replacing human feedback with AI feedback, finding that RLAIF can achieve human-level performance.

LEA's methodology provides the basis for this study. They first conduct SFT on PaLM 2 XS using the well-known Reddit TL;DR dataset to establish baseline model performance and allow the model to grasp task specifics. Then, the authors explored both direct RLAIF, where a separate LLM is prompted for ratings for RL, and distilled RLAIF, where a separate LLM provides soft labels that are used to train a reward model used in RL. They found Direct RLAIF to be superior, with human evaluators favouring its outputs 60% of the time. Additionally, it was demonstrated to outperform SFT. Therefore, our research focuses on building upon the Direct RLAIF approach.

3 Methods

We adapt the methodology laid out in LEA for comparison against the results in NCL. Our experiment is structured as follows.

1. Modelling: Produce three model checkpoints.
   (a) Out-of-the-box;
   (b) SFT; and
   (c) SFT + RLAIF. (This is only applied to the most improved model following SFT.)
2. Evaluation: Compare results between model checkpoints.

In this section, we describe the data in 3.1 , the models selected in 3.2, details on our experimental design described above in 3.3, and the practical

implementation of these models in a low-resource application in 3.4.

3.1 Data

As mentioned, LEA used the Reddit TL;DR dataset which contains a range of topics, some of which may be controversial or inappropriate and therefore introduce biases and undesirable content. Hence, we use the "XSum" dataset prepared by NCL. This dataset consists of news articles from the BBC News website and their accompanying single-sentence summaries. It consists of 226,711 archived BBC articles published between 2010 and 2017. More detail on the preparation of this dataset is available in NCL.

A limitation of this dataset is that there are cases where the summary contains information that cannot be inferred from the text itself. This is addressed in more detail in the discussion (Section 5.3).

In line with our motivation to test low-resource applications, we adjust the approaches of NCL and LEA and use a small subset of the dataset. We extract 10,000 observations and split this into train and test at a 9:1 ratio. For the SFT model checkpoint, we use this split as-is. For the RLAIF checkpoint, we further split the training set, where 20% of the training dataset is used for initial SFT, and the remaining 80% is used for RLAIF. The same test set is used for validation across both model checkpoints. From the extracted observations, we filter out those that exceed the models' maximum input lengths.

3.2 Models

While larger models often achieve impressive results, this study focuses on small-scale LLMs with around 300 million parameters for summarisation tasks: T5-base, GPT-2-Medium, and BART-Large. These models are chosen for their balance between performance, computational efficiency and accessible nature, making them suitable for applications with resource constraints.

T5-base (Raffel et al., 2019) is known for its unique text-to-text unified framework, allowing it to handle a wide range of tasks. It is around 220 million parameters in size and is built on the encoder-decoder transformer architecture similar to that introduced by Vaswani et al. (2017). BART Large also leverages sequence-to-sequence encoder-decoder transformer architecture. BART Large (Lewis et al., 2019) contains around 375 million parameters (BART's size is given relative to the size of BERT. Comparison between the two is made with Lewis et al., 2019; Devlin et al., 2018), and is trained using a unique corrupted text data method. Encoder-decoder models are traditionally considered to be better suited to summarisation tasks as the task of condensing a full document (input text) into a concise summary (output text) aligns better with their architecture. Contrastingly, GPT-2-Medium focuses solely on the decoder portion of the transformer architecture and consists of approximately 345 million parameters (Radford et al., 2019). Decoder-only models perform well in predicting the next word given the input text but are not specifically suited to summarisation tasks.

3.3 Experimental details

Across all checkpoints we make extensive use of Hugging Face libraries such as Transformers[1] and Evaluate[2]. We train models using the T4 GPU available through Google's Colaboratory[3]. This provides a hard limit on the memory we can use and supports our goal to investigate low-resource applications. Details of hyperparameters we use for training and generation are available in Appendices 7.3 and 7.5.

We build the following model checkpoints:

3.3.1 Off-the-shelf

We use the base models to produce summaries of the validation data.

3.3.2 Supervised fine tuning

We conduct fine-tuning on the training dataset by re-training all of the parameters in the pre-trained models. We then use these fine-tuned models to produce summaries of the validation data.

3.3.3 Supervised fine tuning with RLAIF

We first conduct SFT as described above, on the reduced training data, which we use to initialise the policy and value models. Then, we conduct direct RLAIF on the fine-tuned model, similarly to the approach in LEA. However, following experimentation we modify their Direct RLAIF approach by changing the scoring to be from 0.01-0.99 and not processing these scores before feeding them to the RL algorithm. We use Cohere's

[1] Available at https://huggingface.co/docs/transformers/index
[2] Available at https://huggingface.co/docs/evaluate/en/index
[3] https://colab.google/

command nightly model[4] to give a score to the produced summaries which we leverage as the reward for RL. Rather than using the REINFORCE modification in LEA (Williams, 1992), the RL algorithm we used to train the policy is proximal policy optimization (PPO). This method is more conservative and stable than other widely used methods due to the incorporation of techniques like clipping the objective function (John Schulman, 2017). To implement the policy model we use the PPO trainer from the HuggingFace Transformer Reinforcement Learning library (TRL).

Equation 1 shows the loss function of PPO, which combines three elements. The core element is the clipped objective (L_t^{CLIP}), balancing exploration and exploitation. Additionally, PPO incorporates terms for updating the value function (reference network π_θ) and promoting exploration by encouraging diverse actions (entropy bonus ($S[\pi_\theta]$). By optimising this combined loss, PPO trains the agent effectively. For more details, see Appendix 7.4.

$$L_t^{\text{CLIP+VF+S}}(\theta) = \hat{\text{E}}_t[L_t^{\text{CLIP}}(\theta) - c_1 \cdot L_t^{\text{VF}}(\theta) + c_2 \cdot S[\pi_\theta](s_t)] \quad (1)$$

During RLAIF, we monitor the KL divergence to ensure the model does not deviate too far from the original model. For RLAIF training, we use parameter-efficient fine-tuning (PEFT) rather than re-weighting all of the parameters. Whilst performance is typically better in standard fine-tuning, PEFT is less prone to overfitting and is more computationally efficient. Specifically, we implement low-rank adaptation (LoRA), one of the most widely used PEFT methods (Hu et al., 2021). More details are available in Appendix 7.2.

3.3.4 Evaluation

In line with NCL and Dai et al. (2023), we report ROUGE-2 and ROUGE-L statistics, where R-2 measures the overlap of bi-grams between the predicted and reference summaries, and R-L is a Longest Common Subsequence based statistic which takes the longest sequence of words appearing in both texts, regardless of order. The ROUGE metric we report is based on F-1 score. We compare each model checkpoint against the ROUGE score on the XSum dataset of four reference models from NCL and PaLM 2 (Dai et al., 2023): 1) Random, which predicts an article's summary using a randomly selected sentence from the article; 2) Lead, which uses the leading sentence from the article; 3) NCL's **T**opic-aware **Conv**olutional **Seq**uence **to Seq**uence model (T-CONVS2S); and 4) PaLM 2-L, a 340-billion parameter LLM (Elias, 2023).

3.4 Application

Finally, we experiment with the practical implementation of our low-resource summarisation tool, by producing an interactive web app[5] that allows users to interact with our models trained with RLAIF. The memory constraints of this implementation give a practical sense check on the feasibility of implementing our approach in a low-resource environment.

4 Results and Discussion

The ROUGE scores of each model checkpoint and our baselines are presented in Table 1 and Figure 1. The off-the-shelf checkpoint of T5 shows only small improvements over the Random and Lead baselines, BART performs similarly, while GPT-2 performs significantly worse. SFT produces significant improvements relative to off-the-shelf with each model performing close to the best-performing model in NCL (T-CONVS2S). GPT-2 displays the greatest performance improvement from SFT and is selected for RLAIF. Finally, RLAIF on GPT-2 results in a decrease in performance relative to its SFT checkpoint.

Figure 2 displays example summaries produced by each of GPT-2, T5, and BART following SFT. The summaries are generally appropriate, but contain some information that is incorrect or not available from the main text.

Between our models, GPT's superiority over T5 after SFT was unexpected due to the contrasting architectures. As mentioned, encoder-decoder models such as T5 tend to perform better on summarisation tasks. GPT-2's decoder-only architecture is closer to a generative model and less aligned to summarisation. We hypothesise that this may be related to a limitation of the dataset, as discussed below, that makes it an atypical summarisation task.

[4] Hugging Face Card. With thanks to Cohere for generously providing credits under their Cohere For AI Research Grant.

[5] Available at maet-pln-app.streamlit.app, with reproducible code on GitHub.

Model	Checkpoint	R2	RL
Random	Baseline	1.78	11.27
LEAD	Baseline	1.60	11.95
T-CONVS2S	Baseline	11.54	25.75
PaLM 2	Baseline	23.2	N/A
BART	Off-the-shelf	2.9	9.9
BART	SFT	13.4	26.8
T5	Off-the-shelf	3.1	13.5
T5	SFT	11	24.0
GPT-2	Off-the-shelf	0.01	1.3
GPT-2	SFT	11.8	26.2
GPT-2	RLAIF	9.8	23.9

Table 1: ROUGE-2 and ROUGE-L scores for our baseline reference models and each checkpoint of our models. PaLM 2 performs the best, followed by BART following SFT. GPT-2's performance decreased following RLAIF. Random, Lead, and T-CONVS2S scores come from NCL. PaLM 2 scores come from (Dai et al., 2023).

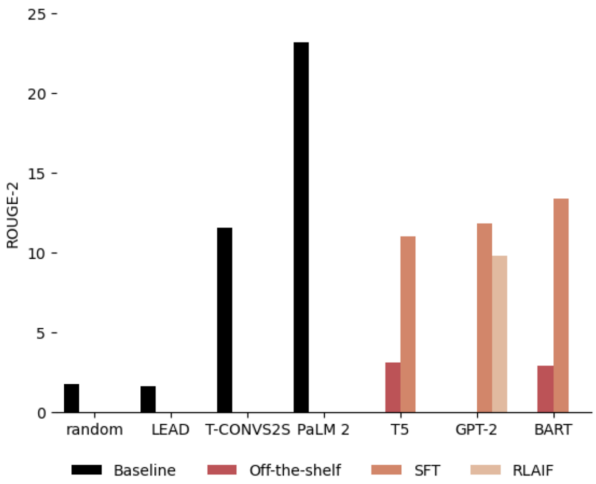

Figure 1: ROUGE-2 scores for our baseline reference models (in black) and each checkpoint of our models (in shades of red). PaLM 2 performs the best, followed by BART following SFT. GPT-2's performance decreased following RLAIF. Random, Lead, and T-CONVS2S scores come from NCL. PaLM 2 scores come from (Dai et al., 2023).

All models perform significantly worse than the PaLM 2-L baseline, which is expected as PaLM 2 is significantly larger than our models (around 1,000 times larger than GPT-2 medium, according to Elias, 2023). This suggests, more generally, that the performance gap between small LLMs and significantly larger LLMs cannot be bridged by SFT and RLAIF, but the difference in performance is encouraging when accounting for the vast difference in model size.

Although the results show that RLAIF did not lead to significant improvements over SFT, anecdotally, we note instances of summaries becoming more consistent with the articles following RLAIF. An example of this is presented in Figure 3. It can be seen that the summary produced by SFT contains false information, but has a higher ROUGE-2 score than the summary produced by RL.

We hypothesise that this discrepancy may be connected to two factors, which we discuss in Section 5.3: 1) the limitations of ROUGE as a N-gram overlap metric, and 2) limitations of the dataset.

5 Conclusions

5.1 Concluding Remarks

In this study, we investigate low-resource SFT and RLAIF on smaller models for text summarisation to increase accessibility to model enhancement. GPT-2, despite having the worst baseline performance (as expected being a decoder-only model) displays the greatest improvements after SFT. However, performance decreases after RLAIF. Hence, we do not find RLAIF to decrease the performance gap with larger LLMs, though find the SFT results promising for this application. We hypothesise this may be a result of limitations concerning the XSum dataset and the ROUGE evaluation metric. Hence, it is suggested RLAIF is performed on smaller models of different architectures, across different datasets and using a greater range of evaluation metrics.

5.2 Future Research

We note that, for the purposes of this study, we limit ourselves to the use of consumer hardware and resources freely available through Google's Colab, specifically the T4 GPU. This raises the question of how much the performance gap could be closed by implementing the same methodology on slightly larger LLMs with higher specification hardware. For example, we suggest that future research explores the performance of LLM checkpoints with closer to one billion parameters. Models of this size would still be dwarfed by our best-performing baseline PaLM 2-L (at a ratio of around 1:300), but are likely to reduce the performance gap further. Additionally, we suggest that future research quantifies the difference in cost and performance of implementing this method-

Summary GPT-2: A farming festival that celebrates Irish heritage, food from Northern Ireland and the best of local products in a north Belfast park is to return to the city for the first time in a decade.

Summary T5: The annual Balmoral Show is being held at the former Maze prison in Ulster, where the best of Northern Ireland's agricultural heritage is showcased.

Summary BART: The 2016 Balmoral Show is getting under way in County Antrim on Wednesday, with more than 100,000 people expected to attend.

Summary Actual: The most important event in Northern Ireland's agricultural calendar - the Balmoral Show - has opened with thousands of people attending.

Article: The three-day extravaganza of farming, food and family fun celebrates many aspects of agricultural life. The Balmoral Show is run by the Royal Ulster Agricultural Society (RUAS) and dates back 148 years. [...] The [former] Maze venue, re-named Balmoral Park, is now hosting the show for the fourth consecutive year. The 2016 event coincides with Northern Ireland's Year of Food and Drink, and local produce features prominently in the exhibitions. [...]

Figure 2: An example of summaries produced by the SFT checkpoint for each of GPT-2, T5, and BART. Text in red shows information that is incorrect or not present in the main text.

Summary SFT: Manchester City's title hopes were ended by a resounding home defeat by Manchester United at the Etihad Stadium.
(ROUGE-2: 10.3)

Summary RLAIF: Manchester City's emphatic 3-0 win at Old Trafford was a mixed bag for David Moyes's side as they were unable to build on their impressive start to the season and now face a fight to even finish in the top six and claim a place in the Europa League.
(ROUGE-2: 8.3)

Article: [...] It was another sobering and thoroughly miserable night for United manager David Moyes as, for the second home league game in succession, the current gap in quality and strength between the reigning champions and their rivals was emphasised as City followed Liverpool by winning 3-0 and with plenty to spare. [...]

Figure 3: An example of a summary produced by SFT (GPT-2) that contains incorrect information but returns a higher ROUGE-2 score than a correct summary after RLAIF. This demonstrates the limitation of ROUGE scores to measure performance in summarisation.

ology on smaller models compared to a larger model. This would enable the application of this research in industry.

Recent advances in RL, such as RL with Augmented Data (Laskin et al., 2020) and self-alignment with limited data (e.g. <100 observations discussed in Guo et al., 2024), may enable further the application of these techniques to specific domains with limited data, and further research should consider combining our methodology with these techniques.

As for all LLMs, our models are prone to hallucinate or omit key information from the summaries. These problems may affect different domains to different degrees. For example, hallucination is more likely to be problematic in financial news summarisation than in creative writing, and omission of key information is more likely to be problematic in legal text summarisation than in other areas. Future research should aim to quantify the risks of these problems after training with our methodology and how they apply to specific domains.

LEA and Bai et al. (2022) evaluated the 'harmlessness' of text generated by state-of-the-art models. This has not been considered in our application to small LLMs. Future research should explore in more detail how this applies to models trained with our techniques, which may be more susceptible to repeating harmful patterns discovered in text.

5.3 Limitations

One limitation of this study is the sole use of the ROUGE metric. Two common pitfalls of N-gram metrics are their inability to reward semantically similar paraphrases and to capture distance dependencies by failing to penalise semantically-critical changes to word order. In line with this, several authors (Paulus et al., 2017; Liu et al., 2016) note that improvements to the ROUGE score do not necessarily correspond with more coherent summaries. We hypothesise that this criticism is relevant here. Specifically, the use of AI feedback encouraged semantically similar summaries without the requirement to increase N-gram overlap.

This limitation could be overcome in future research by using metrics that encourage semantic similarity, such as BERTScore (Zhang et al., 2019) which aims to incorporate semantic similarity by encoding summaries with BERT and measuring their similarity. However, N-gram similarity metrics are well-established and are more widely comparable. By using both N-gram-based metrics and BERTScore, future research would be both broadly comparable and incorporate semantic similarity.

Another hypothesised inhibitor of the perfor-

mance of RLAIF is related to the dataset. There are instances where the true summary contains information that is not available in the article and cannot be inferred, such that the target is not best described as a summary. This happens because the target summaries in the dataset are the sub-title of the article from the website, which are not strictly required to summarise the article and can introduce information that is not discussed elsewhere. An example of this is demonstrated in Figure 4. In this instance, the name, occupation, city, and name of the company owned by the article's subject, which are introduced in the summary, are not included in the article and cannot be inferred.

Anecdotally, we note instances of these cases that the models hallucinate more. In Figure 5 we see a summary that describes a crime and the reward offered for information on the criminal. There are very similar examples in the training dataset where the articles relating to crime then have a target summary including a reward that is not specified previously. We hypothesize the models learn a pattern of, for example, including a reward in the summary if the article is about a crime, instead of learning to summarise the actual content.

We hypothesise that this makes our SFT model checkpoints produce outputs that align better with the dataset target at validation. In contrast, RL enforces actual summarisation which these targets may deviate from. Therefore, RL may be less likely to produce a summary with a high ROUGE score.

Summary: Kemi Adetiba is an award-winning director, producer and cinematographer in the Nigerian city of Lagos and her production company - K-Alpha Innovations - specialises in making music videos.

Article: 22 December 2014 Last updated at 18:15 GMT. A law graduate, she got her first taste of the media industry as a little girl accompanying her father, a radio DJ. As part of the BBC's African Dream series about entrepreneurs, the film-maker explains how she managed to excel in the entertainment industry in Nigeria and across sub-Saharan Africa.

Figure 4: An example of a target summary from the dataset which contains information that is not available in the article. The name, occupation, city, and name of the company owned by the article's subject, which are introduced in the summary, are not included in the article

We perform RLAIF only on GPT-2 as it displays the highest improvement in performance after SFT. Given more computational resources, RLAIF could be performed on all three models concerned as all models performed similarly to T-CONV2S2 after SFT. Hence, a greater scope of comparison between different architectures at each stage would be provided. Furthermore, time constraints mean that we simplify the approach in LEA by using the raw LLM scores without any further processing before passing them to the RL. This may hinder the ability of AI feedback to impact the RL in the same positive way. For the same reason, unlike LEA, we did not explore the probability with which the AI scores are generated. By doing this in future implementations, the confidence and reliability of the AI scorer could be inspected.

Target summary: A reward of up to £5,000 has been offered for information about a sexual assault on a woman in Nottingham.

Summary BART SFT: A £31,000 reward has been offered for information about a woman who was raped in her car.

Article: The attack took place in a car park on Randal Street, Radford. [...] Crimestoppers, the charity which is offering the reward, described the attack as a "very distressing ordeal [...]".

Figure 5: An example of a summary produced by SFT (BART) that contains incorrect information. The summary is very similar to the target summary which also contains information not included in article (shown in red and green, respectively).

6 Code and Reproducibility

To promote the reproducibility of this work, many of the details of this research are shared throughout the paper. Prompting details are elaborated on in Appendix 7.1, parameter-efficient fine-tuning in Appendix 7.2, the RL formulation in Appendix 7.4, and model training and generation details in Appendices 7.3 and 7.5. Please reach out to authors for any additional questions or requests. All of the model checkpoints utilised in this project are publicly available through HuggingFace[6]. All of the code is available on Github[7].

[6] https://huggingface.co/ijwatson98, https://huggingface.co/jth500, https://huggingface.co/reinforcementlucia

[7] https://github.com/jth500/maet-pln

References

Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. 2022. Constitutional ai: Harmlessness from ai feedback. *CoRR*, cs.CL/2212.08073v1.

Jaime Baldeon Suarez, Paloma Martnez, and Jose Luis Martnez. 2020. Combining financial word embeddings and knowledge-based features for financial text summarization UC3M-MC system at FNS-2020. In *Proceedings of the 1st Joint Workshop on Financial Narrative Processing and MultiLing Financial Summarisation*, pages 112–117, Barcelona, Spain (Online). COLING.

Lochan Basyal and Mihir Sanghvi. 2023. Text summarization using large language models: A comparative study of mpt-7b-instruct, falcon-7b-instruct, and openai chat-gpt models. *CoRR*, cs.CL/2310.10449v2.

Sergey Brin and Lawrence Page. 1998. The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*.

Andrew M. Dai, Jonathan H. Clark, Kevin Robinson, Maysam Moussalem, Sebastian Ruder, and Jacob Austin. 2023. Palm 2 technical report. *CoRR*, cs.CL/2305.10403v3.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. BERT: pre-training of deep bidirectional transformers for language understanding. *CoRR*, cs.CL/1810.04805v1.

Jennifer Elias. 2023. Google's palm 2 uses nearly five times more text data than predecessor. https://www.cnbc.com/2023/05/16/googles-palm-2-uses-nearly-five-times-more-text-data-than-predecessor.html. Accessed: 2024-04-07.

Gunes Erkan and Dragomir R. Radev. 2004. Lexrank: Graph-based lexical centrality as salience in text summarization. *Journal Of Artificial Intelligence Research*.

Hongyi Guo, Yuanshun Yao, Wei Shen, Jiaheng Wei, Xiaoying Zhang, Zhaoran Wang, and Yang Liu. 2024. Human-instruction-free llm self-alignment with limited samples. *CoRR*, cs.CL/2401.06785v1.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *CoRR*, cs,CL/2106.09685v2.

Prafulla Dhariwal Alec Radford Oleg Klimov John Schulman, Filip Wolski. 2017. Proximal policy optimization algorithms. *CoRR*, cs.CL/1707.06347v2.

Michael Laskin, Kimin Lee, Adam Stooke, Lerrel Pinto, Pieter Abbeel, and Aravind Srinivas. 2020. Reinforcement learning with augmented data. *CoRR*, cs.CL/2004.14990v5.

Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas Mesnard, Johan Ferret, Kellie Lu, Colton Bishop, Ethan Hall, Victor Carbune, Abhinav Rastogi, and Sushant Prakash. 2023. Rlaif: Scaling reinforcement learning from human feedback with ai feedback. *CoRR*, cs.CL/2309.00267v2.

Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2019. BART: denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *CoRR*, cs.CL/1910.13461v1.

Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries.

Chia-Wei Liu, Ryan Lowe, Iulian Vlad Serban, Michael D. Noseworthy, Laurent Charlin, and Joelle Pineau. 2016. How NOT to evaluate your dialogue system: An empirical study of unsupervised evaluation metrics for dialogue response generation. *CoRR*, cs.CL/1603.08023v2.

Ilya Loshchilov and Frank Hutter. 2017. Fixing weight decay regularization in adam. *CoRR*, cs.CL/1711.05101v1.

Andriy Mulyar and Bridget T. McInnes. 2020. Mt-clinical BERT: scaling clinical information extraction with multitask learning. *CoRR*, cs.CL/2004.10220v1.

Shashi Narayan, Shay B. Cohen, and Mirella Lapata. 2018. Don't give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. *CoRR*, cs.CL/1808.08745v1.

Ani Nenkova and Lucy Vanderwende. 2005. The impact of frequency on summarization.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. *CoRR*, cs.CL/2203.02155v1.

Tatiana Passali, Alexios Gidiotis, Efstathios Chatzikyriakidis, and Grigorios Tsoumakas. 2021. Towards human-centered summarization: A case study on financial news. In *Proceedings of the First Workshop on Bridging Human–Computer Interaction and Natural Language Processing*, pages 21–27, Online. Association for Computational Linguistics.

Romain Paulus, Caiming Xiong, and Richard Socher. 2017. A deep reinforced model for abstractive summarization. *CoRR*, cs.CL/1705.04304v3.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2019. Exploring the limits of transfer learning with a unified text-to-text transformer. *CoRR*, cs.CL/1910.10683v4.

Nathan Schucher, Siva Reddy, and Harm de Vries. 2022. The power of prompt tuning for low-resource semantic parsing. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 148–156, Dublin, Ireland. Association for Computational Linguistics.

Shengli Song, Haitao Huang, and Tongxiao Ruan. 2018. Abstractive text summarization using lstm-cnn based deep learning. *Multimedia Tools and Applications*.

Taylor Sorensen, Joshua Robinson, Christopher Rytting, Alexander Shaw, Kyle Rogers, Alexia Delorey, Mahmoud Khalil, Nancy Fulda, and David Wingate. 2022. An information-theoretic approach to prompt engineering without ground truth labels. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics.

Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. 2022. Learning to summarize from human feedback. *CoRR*, cs.CL/2009.01325v3.

Richard S. Sutton and Andrew G. Barto. 2014. *Reinforcement Learning: An Introduction*, first edition. The MIT Press.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *CoRR*, cs.CL/1706.03762v7.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed H. Chi, Quoc Le, and Denny Zhou. 2022. Chain of thought prompting elicits reasoning in large language models. *CoRR*, cs.CL/2201.11903v6.

Ronald J. Williams. 1992. Simple statistical gradient-following algorithms for connection-

ist reinforcement learning. *Mach. Learn.*, 8(3–4):229–256.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2019. Bertscore: Evaluating text generation with BERT. *CoRR*, cs.CL/1904.09675v3.

7 Appendix

7.1 Prompt Engineering

7.1.1 SFT

Prompt engineering is a rising field that aims to find high-quality prompts that best align language models with the task at hand (Sorensen et al., 2022). The best prompts vary across tasks and models and are especially important to achieve the desired performance in low-resource settings where the models are fine-tuned for the task on a small set of data or even none (zero-shot training) (Schucher et al., 2022).

Therefore, we investigate different templates which vary in complexity during the SFT process. We test all of them on a subset of the data and chose the best-performing subset when training the model on the whole data.

The decoder-only nature of GPT makes it well-suited for next-token prediction but means it needs more detail in the template which guides it to perform the summarisation task. Whereas, for BART and T5, more succinct templates are found to be the best-performing (Figure 6). As has been seen in previous studies, even two semantically invariant prompts generate different outputs (e.g. capitalisation in T5 prompt.).

Moreover, we add beginning-of-sequence (BOS) and end-of-sequence (EOS) tokens to the text to ensure the end and beginning of meaningful content is clear to the models, as GPT-2 and T5 tokenizers do not add these automatically. Adding a BOS after the end of the prompt is key for GPT-2 to generate any text at all.

7.1.2 RLAIF

As described in Section 3.2, we use the Command nightly model from Cohere to produce AI feedback and directly compute a score that we leverage as the reward for RL. We engineer a complex prompt seen in Figure 7. We first explain to the model its role as a scorer and how summaries should be scored in relation to summary quality. Lastly, we employ 0-shot chain-of-thought prompting which was found to maximise alignment to human preferences in Lee et al. (2023) and Wei et al. (2022) and ask the model to give the provided summary a score.

7.2 Parameter efficient fine-tuning: LoRA

LoRA is one of the most widely used PEFT methods. It is a modular approach to fine-tuning in which the pre-trained parameters of the original model are frozen during training and trainable rank decomposition matrices are injected into each layer of the transformer architecture. These are adjusted during training and afterwards merged with the frozen weights to generate the fine-tuned model. This reduces the computational cost massively (Hu et al., 2021).

T5: "summarize: [*article*] "

GPT-2: "You are an expert in text summarisation. You are given the full text. Your job is to summarise the text as concisely and accurately as possible in a single sentence.

TEXT: [*article*]

SUMMARY: "

BART: " Summarize the following document.

[*article*]

Summary:"

Figure 6: The prompts fed to the models during training and evaluation that led to best performance.

7.3 Training hyperparameters

7.3.1 SFT

We initially explore PEFT for SFT but determine it is better to re-train all of the parameters since the small size of our models allowed for it. All of the models are trained for 4 epochs on the training data split. We use the AdamW optimizer with beta1 = 0.9, beta2 = 0.999 and epsilon = 1e-08. This optimizer results in a better training loss and improves generalisation compared to those models trained with Adam (Loshchilov and Hutter, 2017). A linear learning rate schedule is adopted, featuring a 10% warmup phase and a learning rate of 2e-05. Furthermore, we implement mixed precision training, which utilises lower-precision floating-point numbers, accelerating the training process and reducing memory consumption without compromising the accuracy of the models.

7.3.2 RLAIF

After SFT, the policy is updated using LoRA (Appendix 7.2) with a PPO algorithm as mentioned in Section 3.2 and extended in Appendix 7.4. This algorithm leverages the score from Command nightly. The maximum number of tokens allowed for the output of Command nightly is 5 and the temperature 0.2. If no score is found within the response, it defaults to 0.5. For RLAIF, SFT is run for 3 epochs and RLAIF for 1 epoch on a 20/80

split of the training data, respectively.

7.4 PPO

Proximal policy optimization (PPO) is a policy gradient method for RL. Contrasting with standard policy methods, it updates the policy by maximising expected returns of batches of data (while adhering to proximity constraints) rather than updating the gradient at each data sample. PPO methods offer the stability and dependability of trust-region methods, yet they are considerably easier to implement. Only a few lines of code modification to standard policy gradient implementation are needed. PPO is commonly recognised to be a fast and efficient algorithm, often outperforming the more complicated methods (John Schulman, 2017).

The loss equation used is centred around the conservative policy iteration (CPI) loss metric seen in Equation 2. It is an adaptation of the REINFORCE loss. It is the default objective for normal policy gradients which pushes the policy toward actions that yield a high positive advantage over the baseline.

$$\begin{aligned} L^{\text{CPI}}(\theta) &= \hat{\mathrm{E}}_t \left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \cdot \hat{A}_t \right] \\ &= \hat{\mathrm{E}}_t [r_t(\theta) \cdot \hat{A}_t] \end{aligned} \tag{2}$$

In Equation 2, π_θ is a stochastic policy and $\hat{A}_t$ is an estimator of the advantage function at timestep t. $\pi_{\theta_{\text{old}}}$ is the vector of policy parameters before the update.

To apply a proximity constraint the objective is modified to penalise policies that move the $r_t(\theta)$ ratio away from 1. This is shown in Equation 3. This is done by introducing a clipped term. The clipped term contains a truncated $r_t(\theta)$ ratio by applying a clipping operation between 1-ϵ and 1+ϵ (Note ϵ is a hyperparameter). This clipping removes the incentive for moving r_t outside of the interval 1-ϵ to 1+ϵ. Lastly, the minimum operator is applied to achieve the final result. This means whether the advantage is positive or negative, PPO forces policy updates to be conservative if they move very far away from the current policy.

$$L^{\text{CLIP}}(\theta) = \hat{\mathrm{E}}_t \left[\min \left(r_t(\theta) \cdot \hat{A}_t, \text{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right) \cdot \hat{A}_t \right) \right] \tag{3}$$

Equation 3 is then combined with terms for updating the value function (baseline network) and promoting exploration by encouraging diverse actions (entropy bonus). This was seen in 1.

7.5 Generation parameters

For generation of summaries from the articles in validation data, we enable stochastic sampling. The temperature is set to 0.3 and we set the number of beams to 3, to sustain multiple hypotheses of the next token at each step, choosing the sequence with the higher probability. To ensure concise summaries, the maximum number of new tokens is restricted to 150.

Command nightly prompt: " You are an expert in text summarization. Your role is to rate the provided summarization with scores ranging from 0.01 to 0.99, where: 0.01 is the lowest score, 0.99 is the highest score. 0.01 represents a summary that is completely irrelevant, factually incorrect, or incoherent. 0.99 represents a perfect summary that is highly relevant, factually correct, and coherent. The summary should be a single sentence, accurate, and capture the main points of the full text.

Use the following chain-of-thought reasoning to evaluate the summary:

1. Does the summary accurately represent the full text?

2. Is the summary factually correct?

3. Is the summary coherent and easy to understand?

Your response should only be a double precision number that represents the score.

Below, you are given the full text and its summarization.

FULL TEXT: [*article*]

SUMMARIZED TEXT: [*SFT summary*]

Score: "

Figure 7: The prompt fed to the Cohere Command nightly model to generate scores for SFT-produced summaries. [article] is populated with the XSum articles and [SFT summary] refers to the summary to be scored.

10 Peer evaluation form

This form was adapted from one developed at Johns Hopkins University in October 2006; has seen extensive use across a number of universities.

Student number:	23181530	23205704	19006337	20017054	23206473
Attends group meetings regularly and arrives on time	3	3	3	3	3
Contributes meaningfully to group discussions	3	3	3	3	3
Completes group assignments on time	3	3	3	3	3
Prepares work in quality manner	3	3	3	3	3
Demonstrates cooperative and supportive attitude	3	3	3	3	3
Contributes significantly to the success of the project	3	3	3	3	3
Demonstrates energy and enthusiasm	3	3	3	3	3
Total:	21	21	21	21	21

Write *your own* student number in the first, left-most empty column and use the other columns for your fellow group members. Do note that you are expected to provide a self assessment by adding numbers for your own performance as well as for other members. In each column, write a number 0–3 in each column where 3 is "always", 2 is "most of the time", 1 is "sometimes", and 0 is "never" with respect to the statement in the left-most column. Lastly, please sum each column and fill in the total for the last row.